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MILITARY SPECIFICATION

SHELTER, INFLATABLE, ONE SECTION ASSEMBLY (MUST)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a shelter consisting of one semi-cylindrical, air-supported shelter section; two end panels; one frame-supported air lock; one arch support assembly; and related electrical assemblies.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
V-T-285	-	Thread, Polyester
DDD-L-20	-	Label: For Clothing, Equipage, Tentage (General Use)
GGG-H-86	-	Hammer, Hand, (Forged Steel Head)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam
PPP-T-76	-	Tape, Packaging, Paper (For Carton Sealing)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-104	-	Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation-Packaging Methods Of
MIL-P-501	-	Pins, Tent, Metal
MIL-A-3962	-	Anchors, Ground, Arrowhead with Driving Equipment
MIL-C-43285	-	Cloth, Coated (Chloroprene Base Coated, Chlorosulphonated Polyethylene Top Coated)
MIL-S-43766	-	Support Assembly, Arch
MIL-C-43808	-	Cloth, Coated, Nylon

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-202	-	Test Methods for Electronic and Electrical Component parts
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing: with Appropriate Test Methods
MS 35214	-	Screw, Machine - Pan Head, Cross-Recessed, Brass, Black Oxide, UNC-2A

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-4-1262	-	Shelter, Inflatable, One Section Assembly (MUST); Drawing List

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1564 - Testing Slab Flexible Urethane Foam

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Alternate components. Components offered as equivalent to components specified hereinafter and on the referenced drawings as specific manufacturer's part number or equal shall be functionally equal, of equal or better quality, with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of a preproduction sample, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture, as applicable, and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample of the complete shelter to be procured under the specification for first article inspection and approval (see 4.3 and 6.3).

3.3 Initial production testing sample. When specified (see 6.2), the contractor shall furnish a sample for initial production inspection and approval (see 4.4 and 6.4).

3.4 Inspection comparison testing sample. When specified (see 6.2), the contractor shall furnish a sample for comparison inspection and approval (see 4.5 and 6.5).

* 3.5 Materials and components. Materials and components shall conform to the specifications, standards and drawings listed in section 2 and as specified herein (see 6.9).

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* 3.5.1 Cloth, polyester, coated. The coated polyester cloth shall be as specified on the applicable drawing with the following exceptions:

a. That the adhesive used in preparing only the adherability test specimens, per MIL-C-43285, shall be as specified in 3.5.7.
b. That the type I, class 1 coated fabric shall also meet the 200°F (93°C) adherability requirement of MIL-C-43285 with no more than 1/16 inch (2 mm) slippage.

3.5.2 Cloth, nylon, coated. The coated nylon cloth shall conform to MIL-C-43808.

3.5.2.1 Natural or grey. The natural or grey coated, nylon cloth shall conform to type I of MIL-C-43808, with the following exceptions:

a. That the maximum coated cloth weight shall be 8.3 oz/sq yd. (281 g/m^2).
b. That the rectangular swatches of coated cloth, 8 inches by 10 inches (203 by 254 mm) each for the adherability test, shall be cut with the longer dimension parallel to the warp.
c. That the adhesive used in preparing only the adherability test specimens, per MIL-C-43808, shall be as specified in 3.5.7.

3.5.2.2 White. The white coated, nylon cloth shall conform to type II, of MIL-C-43808.

3.5.3 Thread, polyester. All thread shall conform to type I, class 1, subclass B of V-T-285. Dyed thread shall match olive drab shade S-1 (C.A. 66022) and shall show fastness to weathering equal to or better than the standard shade sample. When no standard shade sample is available, the dyed thread shall show good fastness to weathering. Thread sizes shall be as specified on the drawings.

3.5.4 Polyurethane foam. The polyurethane foam shall be a flexible and flame-retardant foam conforming to the following requirements when tested as specified in 4.6.1.2:

Density, lb/cu ft (kg/m^3)	2.6 maximum (41.6)
Indentation load, lb, deflection at 25% $\frac{(N/323\ cm^2)}{(111 + 22)}$	25 ± 5
50% compression set	20% maximum
Tear resistance, lb-force per lineal inch $\frac{(N/m)}{(350)}$	2 minimum (0.9)
After-flame time, seconds	
Original	2 maximum
After oven aging	2 maximum
After steam autoclave	2 maximum

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3.5.5 Overcoating materials. Materials used for overcoating, 3.6.10, shall be air dry coatings meeting color and compositional requirements of the green top coating and pale green top coating of specification MIL-C-43285. When applied to a sample of coated cloth, as used in the shelter, and dried, the resulting sample shall meet the requirements in MIL-C-43285 for blocking resistance and flame resistance.

3.5.6 Dusting compound. The dusting compound shall be whiting, talc, or other finely divided mineral material. The compound shall not support mildew growth. The contractor shall submit a certificate of compliance to the contracting officer that the dusting compound conforms to these requirements (see 4.7).

3.5.7 Adhesive. Adhesives shall be as specified on Drawing 5-4-1694. Containers of the adhesive and the curing agent shall be marked with the manufacturer's batch number and the expiration date and shall be stored in an area where the temperature shall not exceed 70°F (21°C). The batch number identifies that quantity of material manufactured at the same time using the same raw materials or subjected to same unit chemical or physical mixing process intended to make the final product homogeneous. The expiration date, date after which the adhesive or curing agent shall not be used, shall be three months after the date of manufacture. In addition, containers of adhesive and curing agent received in one shipment, shall be identified and segregated to insure their use prior to later deliveries. Before use, each lot of adhesive and curing agent shall be tested for dead load shear strength and dead load peel strength. The adhesive mix shall brush out smoothly without producing drag or showing stringiness. An adhesive lot is approved for use during the 14 days following the date of acceptance (see 4.6.1.4). At the end of 14 day periods, unused adhesive may be retested as long as the expiration date of the adhesive and the curing agent has not been reached.

3.5.7.1 Dead load shear strength. There shall be no more than 1/16 inch (2 mm) slippage at 200°F (93°C), when the adhesive is tested as specified in 4.8.5.1. This requirement is for all adhesives.

3.5.7.2 Dead load peel strength. There shall be no more than 1/2 inch (13 mm) peel at room temperature when the adhesive is tested as specified in 4.8.5.2. This requirement is only for strapping, patching, and splicing adhesive.

3.6 Design and construction. Design and construction of assemblies and components shall conform to the drawings listed on Drawing 5-4-1262 and as specified herein. All visible outside coated cloth surfaces shall be olive green and all visible inside coated cloth surfaces shall be pale green.

3.6.1 Shelter sections. Section assemblies shall be as shown on Drawing 5-4-1264. Each shelter section is of multi-cell construction containing, within each cell, inflatable and replaceable bladders which hold the air to support the section. Normal inflation pressure in each cell of the section shall be 1.50 pounds per square inch (psi) plus or minus 0.25 psi (10.3 $\pm$ 1.7 kPa).

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3.6.1.1 Overpressurization. There shall be no evidence of structural damage such as rupturing of the inner or outer casings, closures, web panels, bladders or manifold assemblies; no edge lifting, tunneling or delamination of strapping or other cemented components; and each pressure relief valve shall show evidence of airflow when each shelter section is tested as specified in 4.8.2.

* 3.6.1.2 Air retention. The internal pressure in each bladder shall be not less than 1.00 psi (6.9 kPa) when each of the bladders of a shelter section is tested for retained internal pressure as specified in 4.8.3.

3.6.1.3 Solar load. There shall be no evidence of edge lifting, slippage, blistering, peeling or other separation of strapping, patch assemblies or other cemented components; tunneling in excess of 1/4 inch (6 mm) width in adhered strapping; delamination of web panel edge reinforcing tapes; or delamination of seams and other cemented components of the bladder assemblies when the shelter section is tested as specified in 4.8.4.

* 3.6.2 End panel assembly. Each end panel assembly shall conform to Drawing 5-4-1275 and be randomly interchangeable with each shelter section assembly by means of the becket loop/fastener tape and end panel support straps when tested as specified in 4.8.1. The end panel assembly contains provisions for attachment of an air lock, air ducts and an electrical utility distribution box.

3.6.3 Stitching, machine. Stitching shall conform to the requirements for stitch types and stitches per inch (stitches/cm) as specified on the applicable drawings and as specified herein and shall be in accordance with FED-STD-751.

3.6.3.1 Thread breaks. Thread breaks shall be overstitched not less than 1 inch (25 mm) at each break on stitch type 301 and not less than 1-1/2 inch (38 mm) at each break on stitch type 401. Thread breaks in type 401 stitching may be overstitched with stitch type 301. Bartacks shall be free of thread breaks.

3.6.3.2 Stitching ends. The ends of type 301 stitching shall be overstitched not less than 1 inch (25 mm), except where ends are turned under in a hem or held down by other stitching or cementing. Where stitching is performed automatically, stitch patterns such as box, box with cross-stitch, "W" stitching or straight line tacking, 3 or more tying, overlapping or backstitches shall be used to secure the ends.

3.6.3.3 Repair of skipped stitches and run-offs. Two or more consecutive skipped stitches or run-off stitches shall be repaired by overstitching. Overstitching shall be accomplished for a distance of not less than 1 inch (25 mm) before, to a minimum of 1 inch (25 mm) beyond, the affected area of skipped stitches or run-offs in type 301 stitching. Skipped stitches or run-offs in type 401 stitching shall be overstitched for a distance of not less than 1-1/2 inch (38 mm) before, to a minimum of 1-1/2 inch (38 mm) beyond, the affected area. Type 301 stitching may be used for overstitching skipped stitches or run-offs in type 401 stitching.

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3.6.3.4 Thread. The addition of any lubricant to the polyester thread prior to or during the sewing operation is prohibited. Water or air spraying for needle cooling is permissible.

3.6.4 Setting of grommets. Holes punched to receive grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommets shall be securely clinched without cutting the materials.

3.6.5 Setting of snap fasteners. If a hole is punched before inserting the male or female part of the fastener, it shall be smaller than the outside diameter of the fastener tube so that the tube must be forced through the hole. The fasteners shall be securely clinched.

3.6.6 Fusing of webbing ends. The cut ends of nylon webbing shall be fused and after fusing, shall be free from frayed edges and loose fibers.

3.6.7 Fusing of polyester and nylon cord. The cut ends of braided cords shall be fused to prevent fraying.

3.6.8 Stitched seams, outer casings. All stitched seams on the outer casings of the inflatable shelter sections shall be strapped as specified on the drawings. Strapping shall be applied with the olive drab side facing out. The strapping shall be adhered across its full width. Strapping shall be applied at each seam in a manner to obtain substantial adhesive contact with both lines of stitching, the areas immediately bordering the stitch lines, and the area in the vicinity of the felled edge. With the shelter section fully inflated, the strapping shall not have any loose edges, unadhered pockets, or wrinkling, and no appreciable longitudinal tunneling at the bottom of web seams, at felled edges or at lines of stitching. Longitudinal tunneling, ie, tunneling parallel to the direction of the web seam, shall be no wider than 1/8 inch (3 mm). There shall be no transverse tunneling i.e., tunneling at an angle to the seam line.

* 3.6.9 Attachment of slide fastener stringers. Care shall be exercised to utilize proper tension during attachment of slide fastener stringers. Slide fastener scoops shall be evenly distributed throughout the entire length of component interface assemblies. The location of separating pins and separating pin retainers shall be maintained as specified on the drawing. Uneven scoop distribution and improper location of pins or retainers is detrimental to the joining capabilities and interchangeability characteristics required. Care shall also be exercised to maintain specified distances between the outer edge of the chain and the stitching, attaching the stringer tapes to the cloth so that there will be no interference with the operation of the slider when mating the stringer halves.

* 3.6.10 Cementing. Cementing of assemblies shall be in accordance with the following procedure:

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a. Surface preparation. Surfaces to be bonded shall be clean and dry prior to application of the adhesive. Coated cloth surfaces to be bonded shall be thoroughly cleaned by rubbing with toluene-wetted clean cloths or pads, which may contain a mild, bonded abrasive of 80, or finer, grit size or may be of a coarse weave (as for scouring). If abrading is exercised, there shall be no scuffing or abrading of the polyester base cloth. Separate clean cloths and clean solvent shall be used for final cleaning wipes. Abrasive coated cloths or pads shall not be used directly on stitching at seams. Foam surfaces to be bonded to coated fabric need not be cleaned unless soiled or dust ladened. Uncoated sides of fabric being bonded (bladders, inner casing patches, etc.) shall be toluene-cleaned, using a clean cloth and clean toluene and allowing the surfaces to dry.

b. Adhesive mixing. The ratio of the adhesive mixture shall be as specified on the drawings. Prior to extracting quantities of the base and accelerator to be mixed, any settlement shall be uniformly redispursed. Containers for mixing, transfer and use shall be clean prior to each filling. Adhesive in the containers shall be covered during transit to work stations and during sustained periods of non-use to prevent loss of solvent by evaporation. After mixing, the adhesive shall be used only while sufficiently spreadable to obtain an even coat. A mix tending to produce drag or showing stringiness when being brushed out has started to cure and must be discarded.

c. Adhesive application. Except for coated cloth-foam laminates, a minimum of two coats of mixed adhesive shall be evenly spread on the cleaned surfaces to be bonded. The first coat shall be dry to the touch before the application of the second coat. Prior to assembly, the adhesive coated surfaces shall not be allowed to be contaminated by a visible layer of dust or other foreign matter. Applied adhesive shall be free of heavy smears or excessively thick lap edges (resulting from insufficient spreading), thin starved areas, uncoated areas or other defects characteristic of improper application. The second coat shall be dried as appropriate for the assembly method to be used; ie, to a tacky (not wet) aggressive grab condition for the two-way, dry stick method; or to a low-tack (dry condition) for two-way stick, solvent reactivation method. Adhesive application on coated cloth-foam laminates shall be either in one coat application to each surface followed by two-way, dry stick assembly, or in one heavy coat application followed be wet-tack assembly.

d. Assembly. Assembly of components shall be made when the tack and grab of the adhesive coated surfaces are in a suitable range and a condition of either excess solvent in the adhesive layers or excessive drying or open time shall be avoided. Surfaces shall be joined in the locations specified on the drawings, applying adequate pressure and rolling or barring firmly to produce an adequate bond. Care shall be exercised to avoid finger contact in the adhesive coated areas along the edges of seams, patches, or other bonded parts. Precautions should be taken to prevent adhesive application, assembling or curing of an assembly at a temperature below 60°F (16°C). The assembled work shall not be moved too soon or in such a way as to result in edge lifting or decreasing strength of bond until the adhesive has set. Surfaces shall be assembled on the same day that they were

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coated, or a third coat of adhesive shall be applied to each surface. When surfaces have been assembled in an incorrect location, the adhesive bond may be immediately separated and reset at the proper location. To facilitate separation, toluene may be applied at the bond line by squirt can or atomizer and the component then peeled away as the adhesive softens. The excess toluene solvent shall be allowed to evaporate before resetting the surfaces. When incorrectly located parts are separated and the adhesive is not in an active condition or has transferred to the other surface, the affected surface shall be given an additional coat of adhesive before reassembling. Any adhesive residue shall not extend further than 1/2 inch (13 mm) beyond the edges of seams, strapping, splices, patches or other bonded parts. Any exposed adhesive within the 1/2 inch (13 mm) allowance shall be overcoated with material specified in 3.5.5. Overcoated areas shall be dusted with the dusting compound specified in 3.5.6. There shall be no unsightly smearing or spreading of adhesive or overcoating and care shall be exercised to insure that surrounding areas are kept clean.

3.6.11 Splicing of cloth. Splicing of inner and outer casing panels of the inflatable section assemblies is prohibited. However, splicing of cloth panels, as specified herein, is permissible. The minimum distance between spliced seams occurring in adjacent panels shall be not less than two feet (610 mm).

3.6.11.1 Bladders. A maximum of two splices per bladder shall be allowed. The ends of pieces shall be cut on a 20° to 45° bias and the shortest length of any spliced piece shall be not less than 4 feet (1.2 m). Pieces shall be joined as specified for the longitudinal lap seams of the bladder as shown on the drawings.

3.6.11.2 Section floor assemblies or section ground cloths. A maximum of four splices per floor assembly or ground cloth, shall be allowed. The shortest length of any spliced piece shall be not less than 4 feet (1.2 m). Pieces shall be joined as specified for panel lap seams as shown on the applicable drawings.

3.6.11.3 Plenum assemblies. A maximum of two splices shall be allowed per inflatable section plenum assembly. The shortest length of piece per plenum assembly shall be not less than two feet (610 mm). Pieces shall be joined as specified for plenum panel lap seam as shown on the drawings.

3.6.11.4 Section containers. A maximum of two splices shall be allowed per section container. The shortest length of any spliced piece shall be not less than three feet (1 m). Pieces shall be joined as specified for the section container panel lap seam as shown on the drawing.

* 3.6.12 Restoration of damages occurring during fabrication. The following components damaged during fabrication, when adjudged repairable shall be repaired in accordance with the requirements specified herein. Cementing shall be performed as specified in 3.6.10. All patches shall have rounded corners.

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3.6.12.1 Bladders.

3.6.12.1.1 Small cuts, tears, punctures, broken yarns. Repairs to small cuts, tears, etc., shall be made by cementing a patch of bladder material to either the coated or uncoated side of the bladder. The patch will be of sufficient size to extend one inch (25 mm) beyond the damaged area. A total of six patches will be allowed per bladder with a maximum area per patch of 18 square inches (116 cm^2). One patch per bladder with a maximum area of 30 square inches (194 cm^2) shall be allowed.

3.6.12.1.2 Missing skim or pin holes. Repairs to missing skim or pin holes shall be made by building up the interior coating with adhesive. Eight such repaired areas shall be allowed per bladder. If the area of damage is of sufficient size it shall be repaired by applying a patch as specified for small holes in 3.6.12.1.1.

3.6.12.1.3 Damaged ring assembly. A damaged ring assembly shall be removed by cutting around the perimeter of the flange and a new ring, prepared in accordance with instructions on Drawing 5-4-1692, shall be installed in the bladder. A collar, cut from bladder material with a 4-1/2 inch (114 mm) ID and an OD of sufficient size to allow a minimum of 1 inch (25 mm) overlap on the bladder shall be cemented to the ring and the outer surface of the bladder with the coated side to the bladder. Only one bladder with this type repair will be allowed per shelter.

3.6.12.2 Inner and outer casing panels.

3.6.12.2.1 Small cuts, holes, tears, punctures, broken yarns, and missing skim. Small cuts, holes, etc., shall be repaired by applying a patch extending not less than 2 inches (51 mm) beyond the damaged area using the same material as specified for the damaged unit. A maximum length of 3-1/2 inches (89 mm) of damage will be allowed for this type repair. A total of four patches with a maximum of 18 square inches (116 cm^2) per patch and a total of two patches with a maximum of 45 square inches (290 cm^2) per patch will be allowed per inner and outer casing.

3.6.12.2.2 Large tears or cuts. Large tears or cuts, with 1/4 inch (6 mm) holes punched at each end of the damaged area to prevent further tearing shall have two patches of the same material as the damaged component cemented one to the inside with at least a 3 inch (76 mm) extension beyond the damaged area and one to the outside with a 2 inch (51 mm) extension beyond the damaged area. The inner patch shall be cemented to the outer casing with neoprene to neoprene or to the inner casing with the coated side of the patch to the uncoated side of the panel. The neoprene side of an outside patch shall be cemented to the hypalon (olive green) side of an outer casing panel. The uncoated side of an outside patch shall be cemented to the coated (pale green) side of an inner casing panel. Not more than one repair of this type shall be allowed per 10 sections.

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3.6.12.2.3 Attachment of webs to inner and outer casing panels. Needle chews in casing panels and webs shall be repaired by cementing a 2 inch (51 mm) wide tape of the same material as the casing panel to the exposed side of the panel and the coated side of the web for reinforcement prior to resewing. Not more than two repairs of this type will be allowed per section with no length limitation.

3.6.12.3 Fabric to foam laminations.

3.6.12.3.1 Cuts, tears, holes. Cuts, tears, and holes shall be repaired by cementing a patch of the same material as the damaged component and extending not less than 2 inches (51 mm) beyond the damaged area. In a complete shelter, a maximum of four patches with a maximum size of 24 square inches (155 cm^2) per patch will be allowed in a fabric to a foam-laminated component assembly. Not more than two patches will be allowed per individual floor assembly or end panel assembly.

3.6.13 Electrical assemblies.

3.6.13.1 Cables. All electrical cable assemblies shall have a DC resistance of less than one ohm when tested for electrical continuity as specified in 4.6.1.3.1. In addition, insulation resistance shall be greater than 100 megohms when tested as specified in 4.6.1.3.2.

3.6.13.2 Distribution boxes. All electrical distribution boxes shall have a DC resistance of less than one ohm when tested for electrical continuity as specified in 4.6.1.3.3.

* 3.7 Anchoring kit. The anchoring kit shall consist of the following components:

1	-	Anchoring Kit container (Drawing 5-4-1451)
34	-	Ground Anchor and Cable; MIL-A-3962, type II, grade B
6	-	Bow Spike Assembly; Drawing 5-4-1570, part number 5-4-1570/1
12	-	Clamp, Manifold; code ident 08484, part number QS200-M32S
2	-	Clamp, Manifold; code ident 08484, part number QS200-M64S
1	-	Head, Driving; MIL-A-3962, for 4 inch (102 mm) anchor
1	-	Hammer, Sledge; GGG-H-86, type X, class 1, steel, 12 lb (5.4 kg)
1	-	Rod, Driving; MIL-A-3962, for 4 inch (102 mm) anchor
1	-	Handle, Holding; MIL-A-3962
22	-	Pin, Tent; MIL-P-501, type II, 12 inches (305 mm) long

* 3.7.1 Repair kit. The repair kit, when required (see 6.2), shall consist of the items specified on Drawing 5-4-1756 and shall be packed in a container conforming to Drawing 5-4-1757.

3.8 Finish of metal components. All metal components shall be finished as specified on the drawings. Finish painted coats shall level out to an adherent, continuous and uniform film without runs, wrinkles, streaks or areas of no film. Any coating damaged during assembly or examination shall be touched up by priming and painting as prescribed on the applicable drawings.

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3.9 Marking.

3.9.1 Nomenclature. The application of the letters "US" and the nomenclature marking on the shelter sections and end panels shall conform to type IV, class 9 of DDD-L-20, in the size characters and locations shown on the drawings. Class 9 markings shall be clearly legible after subjection to accelerated weathering.

3.9.2 Identification. Component assemblies shall be marked for identification in accordance with MIL-STD-130.

3.9.2.1 Parts. Each part shall be marked with its part number, which shall be the same as the drawing number, except the following:

a. Those parts which do not have a suitable or sufficient surface for a part number.
b. Those parts which are permanently assembled shall carry the assembly part number.

3.10 Workmanship. Cloth components shall be free of holes, cuts or tears, thin or weak places caused by abrasion, exposed fabric, blisters, tunnels, unadhered pockets, loose edges or delamination of coating, creased, wrinkles or missing coating. Webbing shall contain no frayed or scalloped edges. Cord and rope components shall be free of cuts, abrasions or breaks. Thread tension shall be maintained so that there will be no loose stitching and seam allowances shall be maintained with seams properly sewed so that no run-offs, pleats or open seams shall result. Care shall be taken in sewing so that there shall be no needle chews. All thread ends shall be trimmed to 1/2 inch (13 mm) or less. Metal components shall be free of burrs, sharp edges, corroded areas and shall not be broken or malformed.

3.10.1 Cementing operation. The cementing operations shall be carried out in such a way as to enable adhesive bonded parts to meet performance and appearance requirements as specified elsewhere in this specification, and to meet the following specific requirements.

3.10.1.1 Adhesion of web seam strapping. When tested as specified in 4.8.6.1, there shall be no peeling of the strapping.

3.10.1.2 Adhesion of round patches on inner casing. When tested as specified in 4.8.6.2, patches shall remain in place on the casing and there shall be no edge lifting at their peripheries.

3.10.2 Welding. Materials to be welded shall be clean and free from scale, corrosion, oil, water and other foreign matter. Welds shall be sound, smooth, free from flash and shall show good fusion.

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3.10.3 Soldering. Materials to be soldered shall be clean and free from scale, corrosion, oil, water and other foreign matter. Soldering shall be adherent and free from spatter and excess solder.

3.11 Durability. The complete shelter shall be capable of withstanding a durability test of 1440 hours without failure when tested as specified in 4.8.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Inspection comparison (see 4.5).
d. Quality conformance inspection (see 4.6).

4.3 First article inspection. When a first article is required, the complete shelter(s) shall be examined for the defects listed in table I, dimensions specified and tested as specified in 4.8.2 and 4.8.3. One shelter section, randomly selected from the first article, shall be tested as specified in 4.8.4. The presence of any visual defect, any dimension not within the specified requirements or failure to meet any test requirement shall be cause for rejection of the first article.

4.4 Initial production sample inspection. When an initial production sample is required, the complete shelter(s) shall be examined for the defects listed in table I, dimensions specified and tested as specified in 4.8.2 and 4.8.3. One shelter section, randomly selected from the initial production sample, shall be tested as specified in 4.8.4. The presence of any visual defect, any dimension not within the specified requirements or failure to meet any test requirement shall be cause for rejection of the initial production sample. In addition, one complete single-section shelter assembly shall be tested as specified in 4.8.9 and any failure shall be cause for rejection of the initial production sample.

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* 4.5 Comparison sample inspection. When a comparison sample is required, the complete shelter(s) shall be examined for the defects listed in table I, dimensions specified and tested as specified in 4.8.2 and 4.8.3. One shelter section, randomly selected from the comparison sample, shall be tested as specified in 4.8.4. The presence of any visual defect, any dimension not within the specified requirements or failure to meet any test requirement shall be cause for rejection of the inspection comparison sample. In addition, one complete single-section shelter assembly shall be tested as specified in 4.8.9 and any failure shall be cause for rejection of the comparison sample.

4.6 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.6.1 Component and material inspection. The contractor shall insure that all components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards and drawings unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.6.1.1 Check valve assembly. The check valve assembly, prior to assembling to the bladder and shelter section shell, shall be tested for compliance with the requirements specified on Drawing 5-4-1685. The sample unit shall be one valve and a lot shall consist of all valves offered for inspection at one time. The inspection level shall be II. Failure of any sample unit (valve) shall be cause for rejection of the lot.

4.6.1.2 Polyurethane foam.

4.6.1.2.1 Foam testing. The polyurethane foam shall be tested as follows for compliance with the requirements of 3.5.4. Any nonconformance shall be cause for rejection of the lot of material.

Characteristic	Test method
Density	ASTM D1564, sec 69-74
Indentation load, deflection	ASTM D1564, sec 19-25
50% Compression set	ASTM D1564, sec 12-18
Tear resistance	ASTM D1564, sec 75-81
After-flame time:	
Original	No. 5905 of FED-STD-191
After oven aging, 22 hrs at 284°F (140°C)	ASTM D1564, sec 45-49.2
After steam autoclave, 5 hrs at 250°F (121°C)	ASTM D1564, sec 5-9.5

4.6.1.2.2 Lot. For purposes of sampling, a lot shall consist of all material produced under like conditions in one unchanged process by one manufacturer and submitted for acceptance at one time.

4.6.1.2.3 Sampling for testing. Sample shall be selected as listed below and subjected to the tests described in 4.6.1.2.1.

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a. Rolls - A sample roll shall be selected at random from the lot. Specimens of sufficient size to conduct the tests shall be taken from the sample after removal of two thicknesses of material from the roll.

b. Sheets - Three bundles shall be selected, the top three sheets of each bundle removed, and sample taken at random from the remainder of each bundle.

4.6.1.3 Electrical assemblies testing.

4.6.1.3.1 Electrical continuity for cable assemblies. Each conductor in each electrical cable assembly shall be tested in accordance with Method 303 of MIL-STD-202. The test current shall not exceed 15 milliamperes. Measurement shall be made between each connector pin which is mutually common to each other electrically. The continuity for each pin shall be checked for compliance with the continuity requirement of 3.6.13.1. Failure to comply shall be cause for rejection of the item.

4.6.1.3.2 Insulation-resistance for cable assemblies. Each cable assembly shall be tested in accordance with Method 302 of MIL-STD-202. The test potential shall be 500 volts, test condition B. Insulation-resistance measurement shall be made between the mutually insulated points and between insulated points and connector shell or ground. Each resistance measurement shall be checked for compliance with the requirement of 3.6.13.1. Failure to comply shall be cause for rejection of the item.

4.6.1.3.3 Electrical continuity of distribution boxes. Each circuit in each electrical distribution box shall be tested in accordance with Method 303 of MIL-STD-202. The test current shall not exceed 15 milliamperes. Measurement shall be made between each terminal pin of the circuit which is electrically common to each other terminal pin. The continuity for each pin shall be checked for compliance with the continuity requirement of 3.6.13.2. Failure to comply shall be cause for rejection of the item.

4.6.1.4 Adhesive testing. Each lot of adhesive shall be tested as specified in 4.8.5.1 for dead load shear strength, and as specified in 4.8.5.2 for dead load peel strength. The inspection lot of adhesive shall consist of all containers of adhesive base from the same batch, and all containers of accelerator from the same batch. The quantities of base and accelerator making up the lot shall be as deemed appropriate for production use within the prescribed 14 day usage limitation as specified in 3.5.7. The sample unit for the dead load shear test shall be one cloth lap seam and two determinations shall be made per sample lap seam. The sample unit for the dead load peel test shall be one cloth laminate and one determination shall be made per sample laminate. Any nonconformance to the requirements of 3.5.7.1 or 3.5.7.2 shall be cause for rejection of the representative lot of adhesive. Containers of the adhesive base and the accelerator for each acceptable lot of adhesive shall be marked with the 14 day usage date and with a lot identity.

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4.6.1.5 Pressure relief valve. Prior to cementing to the bladder, each valve shall be tested in accordance with 4.8.7, for compliance with the pressure relief requirements of Drawing 5-4-1680. Any nonconformance shall be cause for rejection of the valve.

* 4.6.2 In-process inspection.

* 4.6.2.1 In-process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever non-conformance is noted, correction shall be made to the process and all items processed.

a. That the color of all visible exterior and interior surfaces of coated cloth components and assemblies are in compliance with 3.6.
b. That stitching of outer seams, prior to strapping, is in compliance with 3.6.3.
c. That no lubrication is applied to polyester thread prior to or during sewing operations for compliance with 3.6.3.4.
d. That holes punched to receive grommets are in compliance with 3.6.4.
e. That holes punched to receive snap fasteners are in compliance with 3.6.5.
f. That use of the adhesive is in accordance with the cementing instructions specified in 3.6.10.
g. That any fabric splicing and cloth component restoration is in accordance with the requirements of 3.6.11 or 3.6.12, as applicable.
h. That welds and welding are in compliance with 3.10.2 and the drawings.
i. That soldering is in compliance with 3.10.3 and the drawings.
j. That anti-backdraft valve assembly components are chemically treated, prior to priming, and priming is in compliance with Drawing 5-4-1332.
k. That the length of slide fastener stringers, the number of scoops per stringer, the location of stringer pins or retainers, and the distribution of scoops are as specified herein and on drawings.
l. That the 2-1/2 inch (64 mm) bias reinforcing tapes are completely cemented to web assemblies as specified on Drawing 5-4-1690.

4.6.2.2 In-process testing.

4.6.2.2.1 Shelter section. Each complete shelter section assembly shall be tested as specified in 4.8.2 for overpressurization and 4.8.3 for air retention. Any nonconformance with the requirements specified in 3.6.1.1 or 3.6.1.2 shall be cause for rejection of the shelter section. One shelter section, selected at random from a lot of shelter sections shall be tested as specified in 4.8.4 for resistance to solar load. The lot for this inspection shall be fifteen (15) completed shelter sections which have complied with the requirements for over-pressurization and air retention, as specified in 3.6.1.1 and 3.6.1.2, respectively. Any nonconformance with the requirements specified in 3.6.1.3 shall be cause for rejection of the lot of shelter sections.

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4.6.2.2.2 Cementing operation. Each shelter section shall be tested for the adhesion of strapping in accordance with 4.8.6.1. Any failure shall be cause for rejection of the shelter section. One out of every three shelter sections shall be tested for the adhesion of round patches on inner casings in accordance with 4.8.6.2. Any failure shall be cause for rejection of the lot of three sections.

4.6.3 End item inspection. The lot for inspection shall consist of all complete shelters offered for inspection at one time. The sample unit shall be one complete shelter. The inspection sample shall be assembled and inflated to a pressure of 1.5 psi (10 kPa) to facilitate inspection.

4.6.3.1 Visual examination. Visual examination of the complete shelter shall be made for the defects in table I. The inspection level shall be II and the acceptable quality levels (AQLs), expressed in terms of defects per hundred units shall be 40.0 for major defects, 150.0 for major and minor defects combined.

TABLE I. Classification of defects for visual characteristics

Examine	Defect	Classification Major	Minor
Cloth, coated	Hole, cut or tear; thin or weak place due to abrasion; exposed fabric; blister, tunnel or delamination or coating; crease or wrinkle	X	
	Exposed drill hole	X	
	Lump or heavy coated area		X
	Exposed adhesive (not overcoated)	X	
	Failure to dust exposed overcoating	X	
	Excess dusting compound		X
Cloth, cotton duck	Hole, cut or tear; broken or missing yarn	X	
	Multiple floats clearly visible at normal inspection distance (approximately 3 feet (1 m))		X
Webbing	Frayed or scalloped edge	X	
	Cut end not fused		X
Rope, polyester	Cut, broken or abraded yarn	X	
	End not fused		X
Cord, polyester	Cut, broken or abraded yarn		X
	End not fused		X

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Minor
Hardware	Broken	X	
	Malformed	X	
	Corroded area	X	
	Burr or sharp edge	X	
Grommets	Improperly set, two or more teeth exposed		X
	Clinched loosely, allowing grommet to rotate around hole		X
	Clinched excessively tight; i.e., cutting fabric	X	
	Insecurely clinched; i.e., grommets or washers become disengaged	X	
Snap fasteners	Fasteners fail to snap in a closed position or to remain firmly closed	X	
Slide fasteners	Slider jams or fails to interlock chain	X	
	Slider damaged	X	
	Tapes improperly attached	X	
Seams and stitching	Open seam on stitch type 301:		
	- for more than 1 inch (25 mm)	X	
	- for 1 inch (25 mm) or less		X
	Open seam on stitch type 401:		
	- one row of stitching open for 1/2 inch (13 mm) or less		X
	- one row of stitching open for more than 1/2 inch (13 mm)	X	
	- both rows of stitching open (not necessarily at same location)	X	
	NOTE: A seam shall be classified as open when one or more skipped or run-off stitches occur. On double stitched seams, a seam shall be considered open when either one or both sides of a seam are open.		

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitching (cont'd)	Thread breaks:		
	- overstitched less than 1 inch (25 mm) before and beyond break on stitch type 301		X
	- overstitched less than 1-1/2 inches (38 mm) before and beyond break on stitch type 401		X
	NOTE: Thread breaks not overstitched shall be classified as open seams.		
	Skipped stitches or run-offs overstitched less than 1-1/2 inches (38 mm) on stitch type 401		X
	Less than 1 inch (25 mm) on stitch type 301		X
	NOTE: Skipped stitches and run-offs not overstitched shall be classified as open seams.		
	Needle chew resulting in cut, tear or hole	X	
	One or more required rows of stitching omitted (except on boxstitching)	X	
Boxstitching	Wrong stitch type	X	
	Incomplete:		
	- One row of stitching omitted		X
	- two or more rows of stitching omitted	X	
Stitching ends	Not overstitched	X	
	Overstitched less than 1 inch (25 mm) on stitch type 301 (except where ends are held down by other stitching or cementing or turned under in a hem)		X

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Minor
Seams	Pleated		X
Seam type	Wrong seam type	X	
Stitch type	Wrong stitch type	X	
Stitch tension	Loose, resulting in an exposed bobbin or top thread, for more than 6 inches (152 mm)		X
	Tight, as evidenced by puckering on fabric for more than 6 inches (152 mm):		
	- on slide fastener assemblies	X	
	- on all other stitching		X
Stitches per inch	One or more less than specified		X
	Two or more in excess of maximum specified		X
Stitching margin	Larger than specified by more than 3/16 inch (5 mm):		
	- for more than 2 inches (51 mm) but not more than 4 inches (102 mm) in length		X
	- for more than 4 inches (102 mm) in length	X	
Stitching gage	Not as specified		X
Components and assembly	Any component part not fabricated of applicable referenced material	X	
	Any component or assembly with repairs beyond specification allowances	X	
	Any component that will not assemble with its mating component	X	
Reinforcements	Improperly applied (not cemented or stitched as specified), causing excessive fullness on reinforcements or reinforced parts	X	
	Not stitched or cemented to reinforced parts on all sides or not securely caught in seams or stitching of reinforced parts		X

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Classification Minor
Cementing	Puckers, loose edges, unadhered pockets or tunnels	X	
	Excess adhesive in sags or lumps evident through outer layers of fabric	X	
	Exposed adhesive beyond the overcoated area:		
	- on exterior surface		X
	- on interior surface	X	
Strapping	Loose edges, unadhered areas		X
Becket loops	Incorrect number, wrong length, wrong material, not attached as specified	X	
Touch and close fastener tapes	Hook and pile incorrectly positioned, damaged, wrong width, not attached as specified	X	
End wall grommets	Incorrect number, not correctly spaced, damaged, sharp edges, incorrectly attached	X	
Components and subassemblies	Not fabricated of specified material or not in accordance with specified design and specification requirements	X	
Anchoring kit	Component missing; not of specified type, size, or design	X	
Finish of door assembly; anti-backdraft valve assembly	Not finished; not adherent, e.g., blistered, peeled or flaky; coating not continuous or uniform, i.e., not coated or area of no film; scratch, dig or abrasion, exposing bare metal	X	
	Color not as specified		X

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship	Grease or stains	X	
	Thread ends not trimmed to 1/2 inch (13 mm) or less		X
Welding	Missing, incomplete, burnt through, cracked, porous, fractured or otherwise not fused; not type specified	X	
Soldering	Missing, incomplete, not sound	X	
	Flux not removed		X
Marking	Omitted, incorrect, illegible or misplaced		X

4.6.3.2 Dimensional examination. The complete shelter shall be examined for compliance with dimensions specified. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-1 and AQL, expressed in terms of defects per hundred units, shall be 6.5.

* 4.6.4 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully packaged. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE II. Examination of packaging

Examine	Defect
Disassembly	Not as specified
Preservation	Air not expelled
	Components not compactly folded or rolled
	Improper or inadequate cushioning
	Container not as specified
	Excessive void in container
	Closure of container not in accordance with specification

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TABLE II. Examination of packaging (cont'd)

Examine	Defect
Packing	Shipping container not as specified Components susceptible to damage; improperly placed on skid base Excessive void in container Improper or inadequate cushioning, blocking, bracing and anchoring; not in accordance with specified requirements Closure of exterior shipping container not in accordance with specification Strapping or banding omitted, improperly applied or not in accordance with the specification Shipping documents or packing list omitted, incorrect or incomplete
Marking	Omitted; incorrect or illegible Not in accordance with the specification

4.7 Certificate examination. The certificate of compliance for the dusting compound, see 3.5.6, that is submitted as proof of compliance with the specification requirements, shall be examined and validated.

4.8 Tests methods.

4.8.1 Interchangeability - end panel and shelter section assembly. Each end panel assembly shall be assembled randomly to both openings of each shelter section assembly. The end panel and the shelter section assembly conform to the requirements for interchangeability, if assembly can be made without the use of force or damage to either component.

4.8.1.1 Interchangeability retest. The contractor may replace or correct any end panel or shelter section assembly which does not conform to the requirement of 3.6.2. The replaced or repaired component shall be retested as specified in 4.8.1.

4.8.2 Overpressurization, shelter section. The shelter section, with manifold assembly attached and secured to the check manual valves with the clamps (part number QS200-M32S) specified in 3.7, shall be inflated to a minimum pressure of 2.75 psi (19 kPa). This 2.75 psi (19 kPa) pressure shall be maintained for 1/2 hour. At the completion of the test the shelter section shall be examined to determine compliance with the requirements of 3.6.1.1. Any nonconformance shall constitute failure.

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4.8.3 Air retention, shelter section. Each shelter section shall be inflated to a stabilized operating pressure of 1.50 psi (10 kPa) for a period of 1 hour. Each shelter shall remain in the erected mode for a period of not less than 16 hours with the inflation system not in operation. The pressure in each bladder shall be determined after the 16 hour period. The shelter section fails to comply with the requirement of 3.6.1.2 if the internal pressure in any bladder is less than 1.00 psi (6.9 kPa).

4.8.3.1 Air retention, retest, shelter section. The contractor may replace any bladder in the shelter section which does not conform with the requirement of 3.6.1.2 and the shelter section shall be retested for air retention as specified in 4.8.3.

4.8.3.2 Temperature correction. If the ambient temperature changes during the air pressure tests, adjustment shall be made on the basis of a change in pressure of 0.031 psi per degree F (0.21 kPa/°C) change in temperature.

4.8.3.3 Barometric pressure correction. If the barometric pressure changes during air pressure tests, adjustment shall be made on the basis of 0.49 psi (3.38 kPa) per inch of mercury change in barometric pressure. (Correct the internal gage pressure 1 kPa for each 1 kPa change in atmospheric pressure.)

4.8.4 Solar load, shelter section.

4.8.4.1 Apparatus. Apparatus required to perform this inspection shall consist of the following:

a. A test facility utilizing infrared lamps of the incandescent type as the heat source. The test facility shall be capable of subjecting the shelter section to a uniform radiant heat of 200°F (93°C) over the entire outer surface.
b. Instrumentation to measure and monitor the surface temperatures.
c. Water gauge manometers to monitor internal cell pressure.
d. Plastic tubing used to connect manometers.
e. Modified bladder retaining screws. The bladder retaining screws, MS 35214-57 (or brass slotted screws, #10-24UNC-3/4 long) shall be modified by thru drilling a longitudinal hole of sufficient diameter to permit passage of a hypodermic type needle.
f. Hypodermic type needles of sufficient length to provide access to the interior of the bladder.

4.8.4.2 Procedure. Prior to erecting, the shelter section shall be positioned under the test facility. Instrumentation to monitor the temperature shall be affixed to the outer surface of the shelter section at a minimum of four locations, one of which shall be at the top center of the section. Remove one of the bladder retaining screws from each check manual valve assembly and replace with a modified

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screw. At the end of the test, the modified screws shall be removed and replaced with the required screws, applying a small amount of adhesive or other appropriate sealant in the hole prior to replacement of the screw. Following installation of the temperature instrumentation and the modified screws, the section shall be erected and inflated to a pressure of 1.5 psi (10 kPa). A hypodermic type needle shall be inserted through each modified screw and attach the plastic tubing and Manometer gauge. The initial pressure in each cell and the time shall be recorded. The pressure in each cell shall be recorded every half hour and should the pressure in any cell drop to 1.0 psi (6.9 kPa), the shelter section shall be reinflated in 1.5 psi (10 kPa); should the pressure in any cell exceed 2.0 psi (13.8 kPa), the pressure in that cell shall be reduced to 1.5 psi (10 kPa). The shelter section shall be subjected to a solar load which gradually increases the surface temperature, uniformly at all locations, from ambient to 200°F (93°C) plus or minus 5°F (5°C) over a 2-hour period. The 200°F (93°C) temperature shall be then maintained for 4 hours. At the end of this period the temperature shall be gradually decreased to ambient over a 2-hour period. At the completion of the test, the shelter section shall be examined to determine compliance with the requirements of 3.6.1.3. Any nonconformance shall constitute failure.

4.8.5 Adhesive testing.

4.8.5.1 Adhesive dead load shear strength.

4.8.5.1.1 Preparation of sample lap seam. Two rectangular swatches of type I, class 1, MIL-C-43285 coated fabric shall be cut, each approximately 8 inches by 10 inches (203 by 254 mm) with warp threads running parallel to the 10 inch (254 mm) dimension. The swatches shall be cut from areas of the coated fabric that are free of waviness or distortion of the weave. A quantity of the adhesive base and the accelerator, sufficient to fabricate the test samples, shall be selected from the adhesive lot. The two swatches shall be joined with the adhesive to form a lap seam wherein the chlorosulphonated polyethylene side of one swatch is adhered to the chloroprene side of the other. The seam shall be made by overlapping the 8 inch (203 mm) ends for a distance of 2 inches (51 mm), with care exercised that the warp directions of the two swatches are aligned. The surface to be bonded shall be cleaned, adhesive shall be applied, and the assembly shall be made in the same manner used in production (see 3.6.10). Allow the seam to cure at 70 to 75°F (21 to 24°C) for 7 days prior to testing. Adhesives may be tested in less than 7 days curing and if found to pass the requirements will be acceptable.

4.8.5.1.2 Specimens. One-inch (25 mm) wide strips shall be cut from the sample lap seam prepared as specified in 4.8.5.1.1, to provide 1-inch (25 mm) wide by 18-inch (457 mm) long specimens with warp threads running parallel to the 18-inch (457 mm) length.

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4.8.5.1.3 Procedure. One end of each specimen shall be anchored in a suitable grip. A load of 40 pounds (18 kg) shall be fastened to the other end of the specimen. The anchoring grip and the load application grip shall be of a type which precludes failure of the specimen in or near the grips. The grips shall not be applied any closer than 2 inches (51 mm) to the lap seam and shall hold the specimen uniformly across the 1 inch (25 mm) dimension without twist. A suitable gage mark (e.g., ball point line) shall be made at the lap seam edge prior to loading to assist in determining the amount, if any, of relative seam slippage. The two specimens shall be tested for four hours at 200°F (93°C). After removal of load, a second gage mark shall be made at the lap seam edge. The amount of slippage shall be reported as the distance measured to the nearest 1/16 inch (nearest 2 mm) between center lines of the gage marks for compliance with 3.5.7.1. Slippage greater than 1/16 inch (2 mm) shall constitute cause for rejection.

4.8.5.2 Adhesive dead load peel strength.

4.8.5.2.1 Preparation of sample laminate. A sample laminate shall be prepared in the same way as described in 4.8.5.1.1, except that (a) the two pieces of cloth shall have adhesive applied over most of the surfaces of both pieces except for approximately 2 inches (51 mm) at one end and (b) the two pieces shall be assembled in a full overlap in lieu of the 2 inch (51 mm) lap.

4.8.5.2.2 Specimens. One-inch (25 mm) wide strips shall be cut from the laminate. The uncemented end will provide tabs for conduct of the peel test. The tabs shall be further separated by hand for at least 1 inch (25 mm) prior to test.

4.8.5.2.3 Procedure. The specimen shall be subjected to a test load of 1 pound (0.45 kg) (including weight of grip) in dead load peel for 10 minutes at a temperature of 70 to 75°F (21 to 24°C). The anchoring grip and force application grip shall hold the tab ends uniformly across the 1-inch (25 mm) dimension and the force shall be applied uniformly without twist. Before application of the load, a suitable gage mark shall be made at the line of peel on one of the tabs. At the end of 10 minutes, of test, the force shall be removed and a second gage mark made in the same way on the same tab. The distance between the marks shall be measured to the nearest 1/16 inch (nearest 2 mm) as the amount of peel for compliance with 3.5.7.2. Peel greater than 1/2 inch (13 mm) shall be cause for rejection.

4.8.6 Cementing testing.

4.8.6.1 Adhesion of strapping. The test shall be conducted with the shelter section in a deflated condition at a temperature of 70 to 75°F (21 to 24°C). Four web seams straps shall be selected at random for testing. A 6-inch (152 mm) long tab shall be peeled off by hand at any one of the cemented splices in each strap. The area of the section surrounding the tab shall be draped over a table, rod or other horizontal edge so that the strapped area is aligned vertically

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and hangs smoothly. If present, peeled film which has bridged across the peel line and immediate adjoining areas shall be carefully severed by a sharp blade. A test load of 4 pounds (1.8 kg) (including clamp weight) shall be fastened to the tab by means of the clamp which shall hold the tab uniformly and without slippage at a right angle across its width. Within a few seconds after the load is carefully allowed to hang, a gage mark shall be made at the glue line by a ball point writing pen. Ten seconds after loading, a second gage mark shall be made in the same manner. Any visible separation of the two gage marks shall constitute cause for rejection of the section per 3.10.1.1. The tabs of strapping which pass the test shall be recemented back into place.

4.8.6.2 Adhesion of round patches on inner casings. The test shall be conducted with the shelter section inflated to a pressure of 1.5 psi (10 kPa), at a temperature of 70 to 75°F (21 to 24°C). Two light support patches with tape and metal hook and two plasma bottle patches (with braided cord) shall be selected at random for test. A test load of 5 pounds (2.3 kg) pounds shall be suspended from each of the braided cord patches, by hooking into a loop formed by knotting the braided cord, and allowing the load to hang for 2 hours. The test on the two light support patches shall be conducted in the same way except that the test load shall be 10 pounds (4.5 kg) and shall be hooked into the metal loop on the suspension tape. The release of any patch or the edge lifting at the periphery of any patch shall constitute failure of the test and rejection of the lot of sections, per 3.10.1.2.

4.8.7 Pressure relief valve test. Each valve shall be tested utilizing mercury manometers, clamp-in fixtures and some means of detecting airflow through the valve. 2.75 psi (19 kPa) of air pressure shall be applied to the pressure relief valve and flow shall be noted. The 2.75 psi (19 kPa) shall be terminated and then through another circuit, utilizing another mercury manometer or the initial mercury manometer if it is properly sequenced through another circuit, 2.25 psi (16 kPa) air pressure shall be applied and any flow shall be noted. Any flow below 2.25 psi (16 kPa) or non-flow within the range of 2.25 psi (16 kPa) to 2.75 psi (19 kPa) shall be cause for rejection of the valve.

4.8.8 Interchangeability - end panel and shelter section assembly. Each end assembly shall be assembled randomly to both openings of each shelter section assembly of the complete shelter. The end panel and shelter sections assembly conform with the requirements for interchangeability if assembly can be made without the use of force or damage to either component.

4.8.8.1 Interchangeability retest. The contractor may replace or correct any end panel or shelter section assembly in the complete shelter which does not conform with the requirement of 3.6.2. The replaced or repaired component shall be tested as specified in 4.8.8.

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* 4.8.9 Durability test. The shelter shall be tested for conformance with 3.11. The test shall consist of six cycles of 240 hours each for a total of 1440 hours. Any inability to meet the test requirements, as defined below, shall be constituted as a failure. Prior to initiation of the test the shelter shall be packed in a container of approximate dimensions 8 feet by 8 feet by 13 feet (2.4 m by 2.4 m by 4 m) (size of a multi-purpose shelter).

a. The test cycle shall consist of unpacking, assembling, erecting, and connecting all accessories followed by striking the assembly and repacking in the container. The shelter shall remain erect in an operational mode between the time of erecting and striking.

b. A failure is defined as any malfunction which the operator cannot remedy by adjustment, repair, or replacement action within 60 minutes using tools and parts normally available; or which causes, or may cause:

(1) Inability to commence operation.
(2) Cessation of operation during any 240-hour cycle.
(3) Reduction in performance below an acceptance level such as, but not limited to, deflation of more than one bladder; loss of electrical continuity; non-repairable separation of ground cloth from a side wall; non-repairable separation of the tie-down patches or web seams.
(4) Personnel safety hazard due to continued operation.
(5) Serious damage to the shelter or elements thereof due to continued operation.

c. Any malfunction attributable to operator error shall not be considered as an accountable malfunction.

* 5. PACKAGING

* 5.1 Preproduction pack. Unless otherwise specified (see 6.2), a preproduction pack consisting of one completely packaged and marked shelter, shall be required and shall be made available by the contractor at least 15 days prior to starting packaging of the shelters. The preproduction pack shall be examined as specified in 4.6.4 to confirm, prior to starting production, that the preservation, packing and marking comply with the requirements of this specification. Any changes or deviation of preservation, packing and marking of production pack(s) from the approved preproduction pack shall be subject to the approval of the contracting officer. Approval of the preproduction pack shall not relieve the contractor of his obligation to preserve, pack and mark the shelters in accordance with this specification.

* 5.2 Preservation. Preservation shall be level A.

5.2.1 Level A.

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* 5.2.1.1 Disassembly. Air conditioning ducts, manifold assemblies, guy lines (ropes), light fixtures, two power cable assemblies, power distribution cable assembly and electrical receptacles shall be disassembled from the shelter section. The fabric body of the air lock shall be separated from the door assembly end panel. Each bow assembly shall be detached from the air lock body and disassembled into its two components. The light assembly shall be removed from the air locks. The distribution panel and associated power cables shall be disassembled from the end panels.

* 5.2.1.2 Preservation of components.

* 5.2.1.2.1 Shelter section. Each shelter section with the plenum assembly, and section floor assembly each in place shall have the air expelled and flattened. Each flattened section shall be rolled compactly to form a tight roll approximately 144 inches (3658 mm) long and 21 inches (533 mm) in diameter. The roll shall be preserved in the section container assembly.

* 5.2.1.2.2 Support assembly, arch. The arch assembly shall be preserved in accordance with the level A preservation requirements of MIL-S-43766.

* 5.2.1.2.3 Air conditioning duct assembly. The air conditioning duct shall be compressed and restrained by the means provided.

* 5.2.1.2.4 End panel assemblies. Each end panel assembly shall be neatly and compactly folded. A manila paper tag with the following information shall be attached to the door handle. "DO NOT OPEN OR UNLATCH DOORS WHEN ASSEMBLY IS IN FLAT POSITION." The windows, door handles, hinges and other sharp or protruding surfaces shall be cushioned with 1/4-inch (6 mm) thick polypropylene foam cushioning material conforming to PPP-C-1797 secured in place with tape conforming to PPP-T-76.

* 5.2.1.2.5 Air lock assembly. The air lock fabric shall be folded neatly and compactly. The inside and outside ramps of the door assembly shall be folded upward. All metal edges or protruding metal surfaces shall be cushioned as specified in 5.2.1.2.4. The light assembly shall be cushioned with materials as specified in 5.2.1.2.4 and placed in a snug-fitting fiberboard box conforming to PPP-B-636, class, grade and style optional. Closure of the container shall be in accordance with the box specification.

5.2.1.2.6 Bow support assemblies. The open ends and brackets of all the bow assembly components shall be cushioned with material as specified in 5.2.1.2.4.

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* 5.2.1.2.7 Guy lines (ropes). The guy lines (ropes) shall be placed in groups according to the component assembly they are used with. Each assembly group of ropes shall be bundled and securely tied at each end and in the center with cord or twine. Each bundle of ropes shall have attached a manila paper tag with the identification information thereon. The ropes shall then be placed in a close-fitting bag fabricated from polyethylene film conforming to L-P-378, minimum thickness of 0.004 inch (0.10 mm). The bag shall be sealed with tape conforming to PPP-T-76.

* 5.2.1.2.8 Electrical receptacles and power distribution cables. The electrical receptacles and power distribution cables shall be cushioned with material as specified in 5.2.1.2.4. The power distribution cables shall be coiled to a minimum safe diameter.

* 5.2.1.2.9 Electric distribution box. The distribution box shall be packaged in accordance with MIL-P-116, method IIC, and then placed in a PPP-B-636 box, grade V3c, style RSC. The box shall be closed and sealed in accordance with method V, as specified in the appendix of the box specification.

* 5.2.1.2.10 Light assemblies. Four light assemblies shall be unit packed in the storage container provided. The storage container shall be securely closed and packaged in a PPP-B-636 box, grade V3c, and style RSC. The box shall be closed and sealed in accordance with method V, as specified in appendix of the box specification.

* 5.2.1.2.11 Cable assemblies (400 Hz and 60 Hz). Each cable assembly shall have the ends cushioned with material as specified in 5.2.1.2.4.

* 5.2.1.2.12 Manifold assembly and manifold cap assembly. Manifold assembly shall be folded or rolled into compact unit. No unit packing required for manifold cap assembly.

* 5.2.1.2.13 Kits. Each repair kit (when procured) and each anchoring kit shall be cushioned, as required, with material as specified in 5.2.1.2.4, placed in the applicable reusable container provided and unit packed in boxes conforming to PPP-B-636, grade V3c, style RSC. The boxes shall be closed and sealed in accordance with method V, as specified in the appendix of the box specification.

* 5.2.1.2.14 Consolidated unit packing. The preserved air conditioning duct, air lock light assembly, guy lines, electrical receptacles, power distribution cables, two power cable assemblies, manifold assembly and manifold cap assembly shall be consolidated and packed together in a snug-fitting fiberboard box conforming to PPP-B-636, grade V3c, style OSC. Cushioning material, as specified in 5.2.1.2.4 shall be applied as required. The box shall be closed and sealed in accordance with method V, as specified in the appendix of the box specification.

* 5.3 Packing. Packing shall be level A.

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* 5.3.1 Level A packing. Components comprising one complete shelter, preserved, as specified in 5.2, shall be packed in a crate conforming to MIL-C-104, type I, class 2, style a. Care shall be exercised to place and position component items on the skid base in such a manner as to minimize open spaces and to prevent damage of components. Cushioning, blocking, bracing and anchoring of contents shall be provided and shall be in accordance with the appendix of the crate specification and MIL-STD-1186.

* 5.4 Marking. In addition to any special marking required by the contract, bagged and boxed interior unit packs and shipping containers shall be marked in accordance with MIL-STD-129. In addition, packing lists shall be provided as prescribed by MIL-STD-129.

6. NOTES

6.1 Intended use. The shelter covered by this specification is an element of the Field Medical Unit, Self-contained, Transportable (MUST) complex.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Stock Number.
(c) When a first article is required (see 3.2, 4.3 and 6.3).
(d) When an initial production testing sample is required (see 3.3, 4.4 and 6.4).
(e) When an inspection comparison testing sample is required (see 3.4, 4.5 and 6.5).
(f) When a preproduction pack is not required (see 5.1).
(g) When a repair kit is required (see 3.7.1).

6.3 First article. When a first article is required, it shall be inspected and approved under appropriate provisions of ASPR 7-104.55. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Initial production test. When an initial production test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the initial production test sample.

6.5 Inspection comparison test. When a comparison test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the comparison test sample.

6.6 Cementing of coated cloth. The following system has been used in preparing seams where chlorosulphonated polyethylene coated surfaces of coated cloths are to be adhered to chloroprene coated surfaces:

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a. Abrade, by hand, the chlorosulphonated polyethylene surface only to the point of dullness and uniform roughness. Norton Company's Lightning Metalite Cloth - Aluminum Oxide 80-J or 3-M Company's Three-M-ite-Elek-Tro-Cut Cloth J-wt Aluminum Oxide are satisfactory for use in abrading.

b. Wash the abraded chlorosulphonated polyethylene surface and the non-abraded chloroprene surface with clean cotton cloth swabs wet with toluene.

c. Allow each washed surface to dry at room temperature for 15-20 minutes.

d. Brush apply one coat of adhesive (see 3.5.7) to each washed surface.

e. Allow the adhesive to dry at room temperature for 15-20 minutes.

f. Brush apply a second coat of adhesive.

g. After 15-20 minutes drying, reactivate each adhesive with toluene.

h. Brush apply two coats of standard cement to each painted surface, allowing 15-20 minutes drying time between coats.

i. 15-20 minutes after the last coat, reactive each surface, by swab application of reactivating solvent.

j. Laminate immediately. Smooth out by hand, followed by the use of a steel roller.

Note: The standard cement, primer and reactivating solvent referenced above consist of the following:

Standard cement:	Bostik 1095 adhesive - 8 volume Boscodur 9 accelerator - 1 volume
Primer:	Standard cement above - 1 volume Toluene (solvent) - 1 volume
Reactivating solvent:	Standard cement above - 1 volume Toluene (solvent) - 15 volume

* 6.7 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

* 6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.5).

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6.9 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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